



Archive-based multiple feature construction method using adaptive genetic programming

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Abstract

The quality of features is an important factor that affects the classification performance of machine learning algorithms. Feature construction based on Genetic Programming (GP) can automatically create more discriminative features, sometimes greatly improving classification performance. However, constructing a single feature or a small number of features may make the linkage information between labels and features insufficient, resulting in poor classification performance, so we introduce a multi-feature construction method. In addition, premature convergence of the GP may also affect classification performance. This paper proposes an archive-based multiple feature construction method which uses elite archive strategy to preserve and select effective constructed features, and employs an adaptive strategy for GP to adjust the crossover and mutation probabilities based on fitness values. Experiments on ten datasets show that our proposed archive-based multiple feature construction method without using adaptive GP can significantly improve the classification performance compared with traditional single feature construction method, and the classification performance can be maintained or further improved by adding the adaptive strategy. The comparisons with four state-of-the-art techniques show that our proposed method can significantly achieve better classification performance.

Keywords Genetic programming · Feature construction · Adaptive · Multiple features · Classification

1 Introduction

Classification, as a supervised machine learning task, aims to train classification models through training data and classifies unknown instances into corresponding categories based on the trained models [1]. In classification, features are very

important, and the quality of features directly affects the classification performance of machine learning algorithms [2, 3]. However, sometimes original features do not have sufficient discriminative ability, so feature processing for original features is necessary. Feature construction using Genetic Programming (GP) has better interpretability than other feature construction methods such as convolutional neural networks [4], and it can be used to create more discriminative features based on the original features [5] and can achieve very good classification results in some applications, so feature processing is necessary. The effectiveness of GP-based feature construction method has been verified on Balance-scale dataset, Hillvalley dataset and some high-dimensional gene datasets.

Some existing GP-based feature construction methods use GP to evolve an optimal individual, thereby transforming this individual into a single constructed feature [6–8]. However, a single feature constructed by single feature construction methods may contain insufficient information and weak discriminative ability, and affects the classification performance of feature construction. Therefore, the construction of multiple discriminative features has been proposed and the use

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of multi-feature construction method can improve the classification performance of single feature construction [9–11]. Archive-based strategy that preserves top excellent GP individuals during GP runs is a common method for constructing multiple features [9, 11, 12]. Krawiec [12] stored excellent individuals during GP runs to construct multiple features. Ma et al. [11] preserved the solutions produced by multiple generations of GP. Ma and Teng [9] preserved β top excellent individuals during a single GP run and the parameter β is set to 10. However, to decide the number of top excellent GP individuals to be preserved is a problem. To solve this problem, our previous work proposed a two-stage feature construction and feature selection method which first constructs multiple features then selects effective constructed features [2]. This method requires two GP runs and is more time consuming. How to preserve and select effective constructed features, and obtain an optimal subset of constructed features should be further investigated.

As an evolutionary computing algorithm, GP can easily fall into local search which leads to premature convergence [13]. Previous experiments have shown that after several iterations, most of GP individuals have basically the same fitness value and are similar to the locally optimal individuals, which reduces population diversity and prevents GP from searching for better individuals. One reason for causing this problem is to use a fixed crossover rate and mutation rate for all individuals during GP's evolution. Therefore, the self-adaptive mechanism is introduced into the evolution of GP to control the convergence of the population by changing the crossover and mutation rates of individuals using the fitness value, so that the algorithm jumps out of the local optimum and stimulates the population to continue evolving. Adaptive strategy is widely used in particle swarm optimization, genetic algorithm, ant colony optimization, artificial bee colony algorithm, but there are still few adaptive strategy used in GP. Whether adaptive strategy can further enable GP-based multiple feature construction to obtain more effective constructed features and improve classification performance should be investigated further.

1.1 Research goals

The overall goal of this paper is to propose a multiple feature construction method that use elite strategy to preserve multiple excellent individuals, and uses adaptive mechanism to expand the search scope of GP. The following goals are investigated to achieve our overall goal.

Goal 1. Propose an improved multiple feature construction method using elite archive strategy (M_FC) and the adaptive strategy-based feature construction method (A_FC) to verify whether this method can obtain better classification performance than the single-feature construction method (S_FC) and the original features (FULL).

Goal 2. Verify whether our proposed AM_FC can achieve better classification performance than M_FC and A_FC.

Goal 3. Investigate whether our proposed multiple feature construction method using adaptive genetic programming (AM_FC) can achieve better classification performance than four state-of-the-art methods.

1.2 Organization

The remaining sections of this paper are organized as follows. Section 2 provides background information. Section 3 describes our proposed method. Section 4 describes our experimental design. Section 5 presents our experimental results and discussions. Section 6 is our conclusion and future work.

2 Related work

2.1 Research work on feature construction methods

According to different evaluation criteria, we can categorize feature construction methods into three types: embedded methods [14], filter-based methods, and wrapper-based methods. Embedded methods simultaneously construct features and learn a classifier without depending on other classifiers. Embedding methods are simple to use, but a GP classifier itself is often affected by data distribution, which affects the effectiveness of these methods. The filter-based feature construction methods optimize the GP individuals based on intrinsic measures such as Information Gain (IG), Information Gain Ratio (IGR), Correlation, Gini index and Fisher's criterion. Otero et al. [6] proposed a method that uses IGR as a fitness function to measure the quality of constructed features. Muharram et al. [7] proposed a feature construction method based on four fitness functions including IG, Gini index, Chi-square test, and a combination of IG and Gini index. Kamath et al. [15] proposed a filter-based two-stage feature construction method. GP is first used to construct new features and then Genetic Algorithm is used to select informative and effective features. Guo et al. [16] used the Fisher's criterion to extract nonlinear combinations of original features based on the GP. Neshatian et al. [17] proposed a filter-based multiple feature construction method which uses a fitness function that maximizes the purity of the boundary area of classes and needs to run GP multiple times. Wrapper-based feature construction methods are classifier dependent, which use the classification performance of a learning algorithm as the evaluation metric [12]. Wrapper-based methods are usually more time-consuming and achieve better classification performance than filter-based methods. Filter-based methods are faster and their classification performance is usually worse than wrapper-based methods [9].

In general, a GP run will return an optimal GP individual, and the simplest method for feature construction is to construct a single feature. Since a single feature lacks sufficient discriminative information, many multiple feature construction methods [12, 17, 18] have been proposed.

2.2 Research work on multiple feature construction methods

Multiple feature construction methods are proposed to obtain more discriminative features. Neshatian et al. [17] proposed a filter-based multiple feature construction method which constructs the same number of features as the number of classes. The purity of the boundary area of classes is used as a fitness function to evaluate the constructed features. The strategy of coevolution of multiple populations [19] is used to construct multiple features. Other relevant literature on coevolution are shown in [12, 20]. Ahmed et al. [18] constructed multiple features from multiple subtrees of the optimal GP tree and the experimental results show that multiple features generated from subtrees improve the classification performance. Tran et al. [21, 22] constructed multiple features using multiple tree representations and used the terminal nodes of multiple trees as selected features. This method may produce more redundant features. In our previous work, a multiple feature construction method which preserve the optimal individuals produced by multiple generations of GP is proposed [11]. In another multiple feature construction method [9], top excellent GP individuals during a GP run are stored and transformed into multiple features. One of the main problem of our previous work is how to determine how many features to construct. To maintain the classification performance, we construct as many features as possible, which usually contain redundant features.

2.3 Research work on adaptive GP

Adaptive strategies is a method for avoiding local optimum in evolutionary computing algorithms. Srinivas and Patnaik [23] proposed Common adaptive genetic algorithms (AGA) that ensure population diversity by simultaneously modifying the crossover and variance values of individuals. The introduction of adaptive strategies is effective in eliminating problems such as local convergence and preventing algorithmic stagnation. Some other studies have applied adaptive strategies on particle swarm optimization, genetic algorithm, artificial bee colony algorithm, ant colony optimization and so on. There are still few adaptive strategies used in GP. Sanghoun et al. [24] applied an adaptive strategy to GP which changes the crossover and mutation probabilities according to the complexity of the tree structure in each generation. Different from evolutionary programming, which simply uses probability distributions as mutation operators, Hong et al.

[25] designed a hyper-heuristics method with adaptive mutation operators for GP automatically. Eggermont et al. [26] proposed a weighted stepwise adaptation method on GP to solve the symbolic regression problem.

To the best of our knowledge, there is no adaptive strategy used in GP-based feature construction methods. Whether the adaptive strategy can expand the search space of GP and provide more discriminative and effective features for multiple feature construction should be investigated further. Therefore, in this paper, our research will focus on (1) verifying whether our proposed multiple feature construction method (M_FC) and the adaptive strategy-based feature construction method (A_FC) can obtain better classification performance than a single-feature construction method (S_FC) and the original features (FULL), (2) verifying whether our proposed AM_FC can improve the classification performance compared to M_FC and A_FC, (3) investigating whether our proposed multiple feature construction method using adaptive strategy (AM_FC) can achieve better classification performance than other benchmarks.

3 Methodology

The purpose of this paper is to propose an efficient elite archive-based multiple feature construction method AM_FC based on adaptive GP to construct new features and compare it with the existing feature construction methods. The original test sets are transformed into new test sets based on the new constructed features, and used to test the effectiveness of our feature construction method. In this paper, standard GP representation methods are used to address the task of feature construction. GP individuals are represented as a tree structure. The original features are selected as the terminal nodes (leaf nodes) and the internal nodes consist of sets of functions (+, −, ×, ÷). Genetic operators that do not use adaptive strategies, such as crossover, mutation and reproduction follow traditional methods.

3.1 Evaluation metrics - information gain ratio

Information Gain Ratio (IGR) is a splitting criterion of C4.5 decision tree algorithm, which can effectively measure the classification effect of features. IGR is a metric for the filter-based method, where the larger the IGR value of a feature, the more information the feature contains. IGR has a small computational complexity, and can be calculated by reading the training set data once. However, it is only suitable for evaluating a single feature and cannot evaluate multiple features [5, 6]. Therefore, in this paper, IGR is used as a fitness function to evaluate a single constructed feature. The detailed calculation formula for IGR can be found in [9]. In this paper, we use the wrapper-based approach to evaluate the constructed

multiple features by using the classification performance of the learning algorithm in M_FC and AM_FC.

3.2 Overview of feature construction methods based on adaptive strategy

The algorithm will gradually converge as GP runs, that is, the fitness value of GP individuals gradually increases with iteration until it becomes stable. However, problems such as premature convergence, local optimum and stagnation evolution [27] will occur, which will affect GP's convergence speed and search space. Therefore, in this paper, we don't follow the traditional fixed crossover and mutation probabilities in genetic operators any more, but instead adopt an adaptive strategy that combines the individual's crossover and mutation probability with their fitness values [23]. The motivation for using this strategy is to prevent GP from falling into local search, and enrich the diversity of the population [28]. Experimental results in Sect. 5 have verified the effectiveness of this strategy.

The larger IGR, the greater individual's fitness value, indicating that the individual is better. The calculation formulas for crossover and mutation probabilities are shown in Eqs. 1 and 2. In Eq. 1, the crossover probability of an individual is related to the larger fitness value of two crossover individuals f' , as well as to the maximum fitness value of the population f_{max} and the average fitness value of the population f_{avg} where $f_{max} - f_{avg}$ denotes the degree of population convergence. The smaller $f_{max} - f_{avg}$, the more degree tends to converge of the population. In order to jump out of local optimum, it is necessary to increase the probability of crossover and mutation probability, generate more individuals, and expand the search range of GP in the population. Therefore, the denominator of $f_{max} - f_{avg}$ in Eqs. 1 and 2 is inversely proportional to the convergence degree of the population. However, finding the optimal individual is the key to solve our feature construction problem. In order to preserve excellent individuals, based on Eqs. 1 and 2, individuals with higher fitness values than the average fitness value have a lower crossover and mutation probability. In Eq. 1, individuals below the average fitness value must participate in crossover to eliminate these individuals and produce more promising and superior individuals. We set $k_1 = k_2 = 1$, which is consistent with literature [23]. In Eq. 2, the parameters k_3 and k_4 are used to guarantee that random mutations will not be increased, but also allows the offsprings to inherit the good characteristics of the original individual. In the literature [23], $k_3 = k_4 = 0.5$. But through experiments, we found that excessively high k_3 and k_4 values could destroy promising individuals and make GP unstable. A large number of experiments have shown that setting k_3 and k_4 to 0.2 is more appropriate.

If the best individual don't participate in the crossover and mutation operation, it will lead to the exponential rapid reproduction of these individuals in the population, which results in premature maturity. So the crossover probability of the best individual (P_{cb}) is set to 0.1, and the mutation probability of the best individual (P_{mb}) is set to 0.02, which gives the best individual a chance to participate in future evolution. We have verified the effectiveness of parameter settings of crossover and mutation probability of the best individual through experiments. A detailed discussion of the corresponding parameter setting is shown in Sect. 1.1. Compared to GP without using the adaptive strategy, the population of adaptive GP will have greater diversity, and have a higher probability of producing the optimal individual, which is conducive to expanding the search space of GP and finding the global optimal solution.

$$P_c = \begin{cases} k_1 \frac{f_{max} - f'}{f_{max} - f_{avg}}, & f' \geq f_{avg} \\ k_2, & f' < f_{avg} \end{cases} \quad (1)$$

$$P_m = \begin{cases} k_3 \frac{f_{max} - f}{f_{max} - f_{avg}}, & f \geq f_{avg} \\ k_4, & f < f_{avg} \end{cases} \quad (2)$$

where P_c is the crossover probability, P_m is the mutation probability, f_{max} represents the highest fitness value in the population, f_{avg} represents the average fitness value in the population, f' is the the larger fitness value of the two crossover individuals, f represents the fitness value of mutation individual. k_1, k_2, k_3 and k_4 is the control parameters. In this paper, $k_1 = k_2 = 1$ and $k_3 = k_4 = 0.2$.

3.3 Elite archive-based multiple feature construction method

As GP evolves, the excellent individuals generated during one generation may be lost. Once these individuals are lost, they will be difficult to rediscover in later evolutionary processes. Therefore, in this paper, an elite archive strategy is proposed to preserve multiple individuals generated in GP process. All top excellent individuals generated during GP evolution are preserved and stored in an archive. In each generation, individuals in the archive need to be compared with individuals in the population, and individuals with lower fitness values in the archive are replaced by individuals with higher fitness values in the population, in order to ensure that individuals in the archive have the highest fitness values in each generation.

If all the individuals of the archive are transformed into constructed features, there will be a large number of redundant features. So in this paper, a forward search method is used to select effective subsets of constructed features. A

set φ is used to gradually add individuals selected from the archive. At the beginning, the set φ is set to empty. Individuals are selected according to the order of fitness values in the archive from the largest to the smallest, and incrementally add the individuals to the set φ each time, and form $\Phi = \{\varphi_1, \varphi_2, \dots, \varphi_i, \dots, \varphi_\beta\}$, where β is the number of individuals in the archive, i ranges from 1 to n , and the corresponding subscript of each set denotes the number of individuals in the set φ . Then, the set φ are tested one by one by the classifier and the individuals in the set φ_i with the highest classification accuracy in Φ are selected as our selected individuals, where the $\varphi_i = \{I_1, I_2, \dots, I_i\}$. In the AM_FC method, we have conducted experiments using archive sizes of 100, 200, 300, 400. When n is taken as 100 or 200, there is no difference in the final classification results, but the experimental results are better than that when n is taken as 300 or 400. Therefore, in order to reduce the computational complexity, $n=100$ is adopted in our experiments. Our proposed multiple feature construction method based on adaptive genetic programming is shown in Fig. 1.

3.4 Variants of the algorithms

For the convenience of experimental comparison in Sect. 5, we define different algorithm variants based on whether to use elite archive and adaptive GP. The algorithm variants are described as follows.

FULL: This algorithm uses original features that are not processed for classification directly.

S_FC: This algorithm is a single feature construction method to preliminarily verify the effectiveness of the feature construction method.

A_FC: This algorithm is a single feature construction method based on an adaptive strategy, aimed at verifying the effectiveness of the adaptive strategy.

M_FC: This algorithm employs standard GP and uses our elite archive strategy to construct multiple features.

AM_FC: This algorithm is an elite archive-based multiple feature construction method based on adaptive GP. The pseudo-code of the AM_FC algorithm is shown in Algorithm 1.

4 Experiments

4.1 Benchmark techniques

In order to verify the effectiveness of our proposed method AM_FC, four state-of-the-art techniques are selected for comparison. The first is a filter-based multiple feature construction method proposed by Neshatian et al. [17]. This method proposes a fitness function that maximizes the purity of the boundary area of classes and needs to run GP multiple

Algorithm 1: Pseudo-Code of AM_FC algorithm

Input: $\mathcal{D}$: Divided training set
Output: the constructed individuals

- 1 Randomly initialize the individuals in the population of GP
- 2 Initialize the population P with size p and storage pool Φ with size β
- 3 **while** *Maximum Generations is not reached* **do**
- 4 Evaluate individuals fit on the training set $\mathcal{D}$ according to the fitness function
- 5 Selecte individuals from the population using the Tournament method
- 6 Change the crossover and mutation probabilities between individuals according to the adaptive strategy in 3.2
- 7 Remove duplicate individuals
- 8 **for** $i = 1$ to p **do**
- 9 **for** $j = 1$ to β **do**
- 10 **if** $fit_j < fit_i$ **then**
- 11 Insert i to the archive and replace j in the archive.
 //update the archive Φ_k
- 12 **end**
- 13 **end**
- 14 **end**
- 15 **end**
- 16 The β features constructed in the archive Φ are obtained.
- 17 According to the multiple feature construction strategy in 3.3, the constructed feature set $\mathcal{I}_d = \{I_1, I_2, \dots, I_d\}$ is confirmed.
- 18 Return the constructed individuals

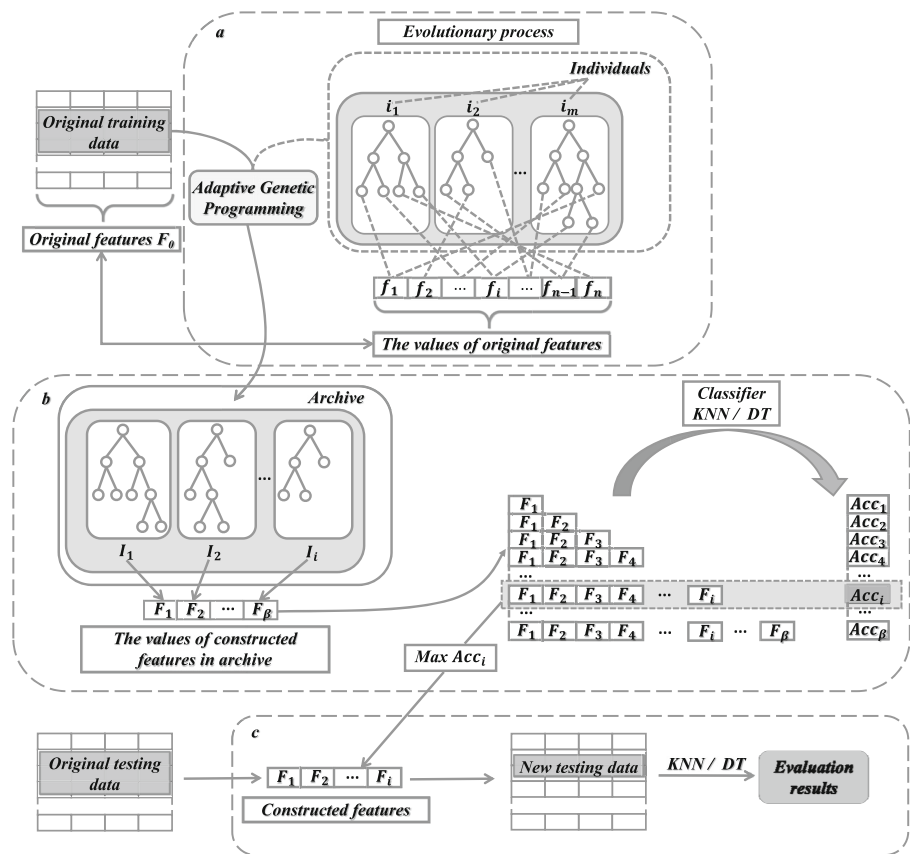
times. This method is named as FcNesh. The second method was proposed by Tran et al. [29], which combines selected features (leaf nodes) with constructed features, and performs feature selection and feature construction at same time. This method is named as FcTran. The third method is proposed by Ma et al. [9]. This method uses a hybrid fitness function, and stores top excellent individuals during a single GP run. This method is named as Fma. The fourth method is proposed by Ma et al. [11]. This method uses multiple evaluation criteria and elite archive strategy to obtain optimal individuals generated by multiple generations. This method is named as FcMa.

4.2 Experimental design

Our proposed methods were evaluated on ten datasets, and the detailed description of the datasets is shown in Table 1. The first five datasets were collected from UCI Machine Learning Repository [30]. The last five datasets involving gene microarray [31, 32] were obtained from <http://csse.szu.edu.cn/staff/zhuzx/datasets.html> [33]. The datasets have different numbers of features, instances, and classes, which are representative examples for solving classification problems and can be used for verifying whether our constructed features can achieve good classification effectiveness.

GP runs on Python-based DEAP framework, and the parameter settings of GP are shown in Table 2. The feature sets in GP are +, −, ×, ÷. When the operator ÷ encounters

Fig. 1 The Overview of our proposed AM_FC method. (Part a denotes the evolutionary process using an adaptive strategy; In part b, the archive-based strategy is used to identify the elite individuals and the forward selection mechanism is applied to select constructed features from the elite individuals; Part c denotes transforming the testing data using the constructed features and performing classification)



a denominator of 0, it returns 0. The terminal set of GP is the original features. The other parameters of GP are consistent with the literature [11], and when compared with other papers, they are consistent with their papers' parameters [9, 17, 29].

To make the experimental results more convincing, we randomly divided the dataset into 30 different training and test sets using thirty different random seeds [34], and used 70% of the dataset as the training set and the remaining 30% as the test set [35]. The training sets are used to construct new individuals, and the testing sets generate a new testing sets through the transformation of the new individuals, and evaluates the classification performance of the new testing sets. 10-fold cross-validation is used to evaluate the classification performance. Classification accuracy is used as the evaluation metric. GP runs independently 30 times on 30 training and testing sets, and the average accuracy of the 30 GP runs is used as the evaluation result to avoid the impact of dataset partitions and stochastic characteristics of GP on experimental results. From sections 5.1 to 5.3 of the experiment, K-nearest Neighbors (KNN, $K = 5$) and Decision Tree (DT) are used to evaluate the effect of the feature construction.

Table 1 Description of the ten datasets

Dataset	#Features	#Instances	#Classes
sonar	60	208	2
wdbc	30	569	2
iris	4	150	4
liver-disorders	6	345	2
WBC	9	683	2
Leukemia	7129	72	2
Lymphoma	4026	62	3
MLL	12582	72	3
Ovarian	15154	253	2
Colon	2000	60	2

5 Experimental results and discussions

To verify our research goal, the following experiments were arranged. (1) The feature construction method (S_FC), which used information gain ratio (IGR) as the fitness function, was compared with the original feature FULL to verify the effectiveness of the feature construction method. (2) Comparing our proposed M_FC with S_FC and FULL to verify whether our proposed multiple feature construction method with elite archive is effective. (3) Comparing our proposed A_FC

Table 2 Parameter settings of GP

Parameters	Parameter value
Function set	+, −, ×, ÷ (protected division)
Terminal set	Original features
Population size	500
Number of generations	50
Initialization	Ramped half-and-half
Mutation method	Random subtree creation
Selection method	Tournament method(size = 7)
Maximum tree depth	17
Starting mutation probability	20%
Starting cross-over probability	80%

method with S_FC and FULL to verify the effectiveness of our feature construction method with an adaptive strategy. (4) Comparing our proposed AM_FC method with FULL, S_FC, A_FC and M_FC methods to verify the effectiveness of the method combining archive and adaptive strategies. (5) Comparing our proposed AM_FC method with other benchmark techniques to verify the effectiveness of our AM_FC method.

5.1 The estimation of the parameters P_{cb} and P_{mb}

The parameters P_{cb} and P_{mb} are used to adjust the crossover and mutation probabilities of the best individuals during the evolutionary process. To verify the impact of the parameters P_{cb} and P_{mb} on the experimental results, we have done a large number of experiments. In the experiments, we first set $P_{cb} = 0.1$ and $P_{mb} = 0.01$, then gradually increase the value of the parameters. Figure 2 shows the classification accuracy of sonar and Colon datasets under different values of P_{cb} and P_{mb} in the AM_FC method. As shown in Fig. 2, the accuracy decreases as P_{cb} increases, while the accuracy first increases and then decreases as the value of parameter P_{mb} increases. The highest classification accuracy can be obtained when $P_{cb} = 0.1$ and $P_{mb} = 0.02$, and this results can be similarly got in other datasets. As the crossover and mutation probabilities increase, the probability of destroying the best individuals also increases, which is not conducive to protecting high-quality individuals. Only appropriate P_{cb} and P_{mb} can solve the problems of stagnation or local optima, which protects excellent individuals while moderately participating in the exploration of the population. Thus, we set the crossover probability of the best individual $P_{cb} = 0.1$ and the mutation probability of the best individual $P_{mb} = 0.02$.

5.2 Comparisons of five feature processing methods

In this section, four different feature processing methods including the baseline methods FULL and S_FC, the fea-

ture construction method based on adaptive strategy A_FC, the feature construction method with elite archived strategy M_FC, the feature construction method with adaptive and elite archived strategy AM_FC are compared. The experimental results are shown in Table 3. In the table, “Wtest” denotes Wilcoxon significance test with a significance level of 0.05. “+” or “−” denotes that the experimental result of the AM_FC is significantly better or worse than other methods. “=” denotes that the experimental result of one certain method is similar to the result of the AM_FC method. “++” indicates the experimental results of AM_FC are significantly better than other methods with p -value = 0.01. In addition, another non-parametric test, the Friedman test, is used to verify whether there is a significant difference among feature construction methods and marked on boxplots with green or yellow color to indicate significant performance differences (p -value < = 0.05) or borderline significant differences ($0.05 < p$ -value < = 0.4) between the Post-hoc analysis for Friedman test of the Friedman test [36]. The boxplots of performance differences in KNN classifier and DT classifier are shown in Figs. 3 and 4.

5.2.1 Comparisons between FULL and S_FC

S_FC is a single feature construction method. To verify the effectiveness of the basic feature construction method, S_FC is compared with FULL. In Table 3, we can see that compared to the original dataset FULL, S_FC has improved performance on 6 datasets, reduced performance on 2 datasets and maintained performance on 2 dataset. For KNN classifier, the classification accuracy of S_FC is 1.13%, 3.58%, 7.74%, 5.02%, 10.83% and 3.43% higher than that of FULL on datasets wdbc, liver-disorders, Leukemia, MLL, Ovarian and Colon. For DT classifier, the classification accuracy of S_FC is 4.39%, 0.41%, 2.64%, 17.39% and 6.11% higher than that of FULL on datasets Lymphoma, Ovarian, wdbc, Leukemia and MLL, respectively. The experimental results show that the basic single feature construction method can improve

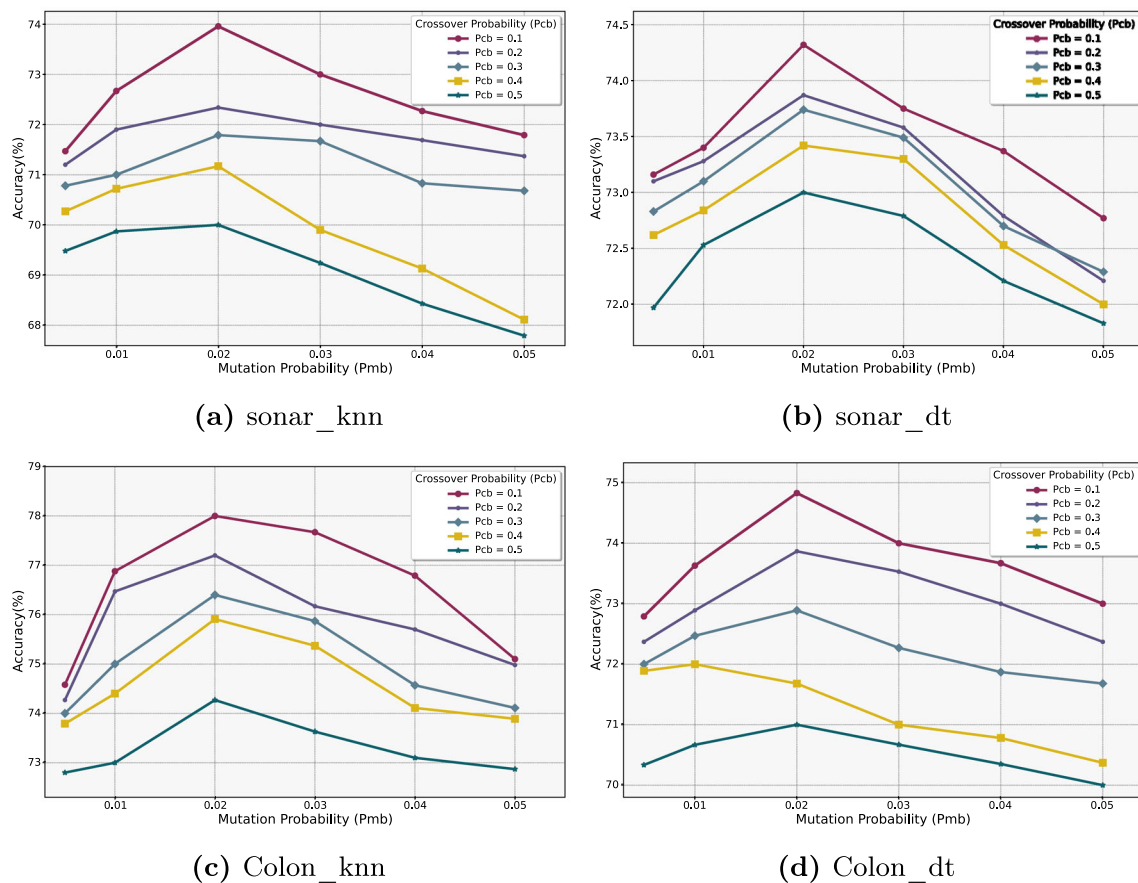


Fig. 2 The impact of P_{cb} and P_{mb} on the classification results

the classification performance on most datasets compared to FULL. However, a single constructed feature contains less discriminative information and still cannot obtain satisfactory results. In the following sections, the effectiveness of our improved multiple feature construction method is shown and analyzed.

5.2.2 Comparisons between S_FC and M_FC

Our proposed M_FC used an elite archive strategy to construct multiple features. To verify the effectiveness of our proposed M_FC and S_FC are compared with the method. As shown in Table 3, compared with S_FC, M_FC method can obtain higher classification results on 9 out of 10 datasets of KNN, and 9 datasets can perform better on DT. The classification accuracy of M_FC is 11.32%, 11.63%, 6.2%, 8.33% and 7.83% higher than that of S_FC on datasets sonar, iris, live-disorders, MLL and Colon in KNN classifier. On DT classifier, compared with S_FC, the classification accuracy of M_FC has increased by 6.84%, 3.1%, 2.16% and 7.5% on datasets Lymphoma, liver-disorders, Leukemia and MLL, respectively. Moreover, boxplots marked by Friedman test in Fig. 3 and Fig. 4 show that M_FC is significantly better

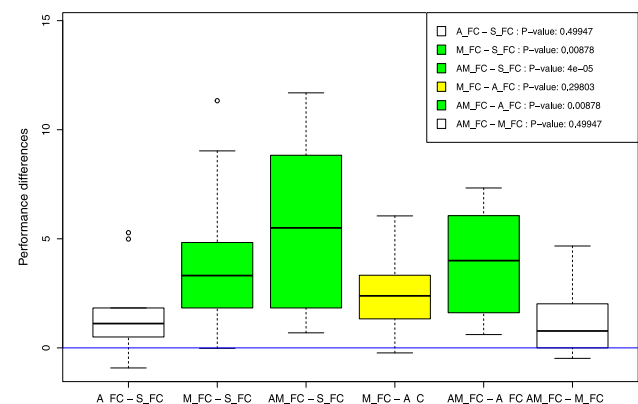
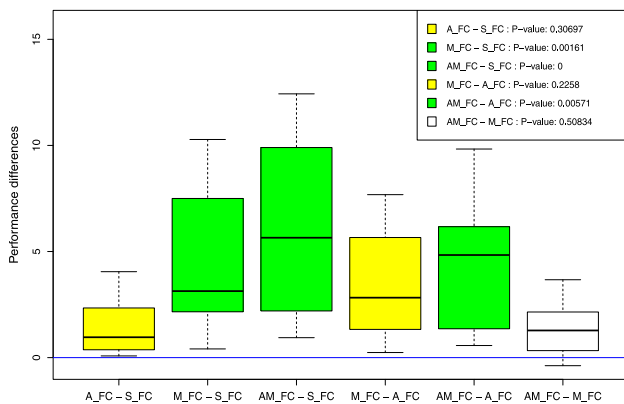


Fig. 3 Boxplot of performance differences between paired feature construction methods (S_FC, A_FC, M_FC and AM_FC) under KNN classifier, marked by Post-hoc analysis for Friedman test (P-value = $2.285e-06$)

than S_FC with P-value=0.00878 in KNN classifier and is significantly better than S_FC with P-value = 0.00161 in DT classifier. The experimental results show that our multiple feature construction method has achieved significant improvement compared to the basic single feature construc-

Table 3 The experimental results of four feature processing methods

Dataset	Method	#F1	A±Std-KNN	Wtest-KNN	#F2	A±Std-DT	Wtest-DT
sonar	FULL	208	68.47 ± 5.40	++	208	68.72 ± 10.01	++
	S_FC	1	63.28 ± 10.80	++	1	61.53 ± 8.99	++
	A_FC	1	68.27 ± 6.65	++	1	64.13 ± 5.86	++
	M_FC	11.47	72.31 ± 6.71	+	15.27	71.81 ± 7.71	+
	AM_FC	20.87	74.32 ± 5.42		43.93	73.96 ± 4.32	
wdbc	FULL	30	95.26 ± 1.49	+	30	92.22 ± 2.05	+
	S_FC	1	96.39 ± 1.57	+	1	94.86 ± 2.07	+
	A_FC	1	95.47 ± 2.09	=	1	95.23 ± 2.44	=
	M_FC	10.53	96.37 ± 1.58	+	25.23	95.47 ± 2.29	=
	AM_FC	10.07	97.08 ± 1.30		32.77	95.80 ± 2.05	
iris	FULL	150	94.79 ± 2.82	+	150	93.48 ± 3.80	+
	S_FC	1	84.02 ± 12.34	+	1	85.78 ± 12.59	+
	A_FC	1	89.30 ± 5.74	++	1	89.83 ± 5.69	+
	M_FC	12.17	95.35 ± 4.18	=	24.77	95.52 ± 2.91	=
	AM_FC	11.4	95.72 ± 3.54		16.57	95.68 ± 2.96	
liver-disorders	FULL	345	59.13 ± 5.32	++	345	59.77 ± 6.65	+
	S_FC	1	62.71 ± 7.70	++	1	59.52 ± 6.74	+
	A_FC	1	63.77 ± 6.16	++	1	59.64 ± 6.58	+
	M_FC	25.07	65.18 ± 5.88	+	32.97	62.62 ± 5.20	=
	AM_FC	30.8	69.74 ± 5.71		40.43	64.15 ± 5.79	
WBC	FULL	9	96.38 ± 1.25	+	9	94.19 ± 1.82	+
	S_FC	1	95.54 ± 1.66	=	1	94.13 ± 1.86	+
	A_FC	1	96.25 ± 1.48	=	1	94.21 ± 1.67	+
	M_FC	13.03	96.02 ± 1.08	=	36.17	94.54 ± 1.89	=
	AM_FC	22.5	96.86 ± 1.17		28.30	95.57 ± 1.55	

**Fig. 4** Boxplot of performance differences between paired feature construction methods (S_FC, A_FC, M_FC and AM_FC) under DT classifier, marked by Post-hoc analysis for Friedman test (P-value = 2.426e−05)

tion method, which proves that our elite archive and selection strategy is effective.

5.2.3 Comparisons between S_FC and A_FC

To verify the effectiveness of the A_FC method using an adaptive strategy, we compared it with the S_FC method. In KNN classifier, the accuracy of A_FC is higher than FULL on 6 datasets, especially on MLL, Ovarian and Colon datasets, which are 6.85%, 11.06%, and 4.39% higher, respectively. Similarly, in DT classifier, the accuracy of A_FC method is higher than FULL on 7 datasets, especially on iris, Leukemia, Lymphoma, MLL datasets, which are 4.05%, 18.22%, 6.73%, and 7.95% higher respectively. As shown in Fig. 3 and Fig. 4, though A_FC can not achieve significantly better classification performance than S_FC, the experimental results in Table 3 show that A_FC can obtain higher classification performance than S_FC in almost all datasets. To demonstrate why our adaptive strategy can improve the classification performance, we drew scatter plots of population diversity for A_FC and S_FC on the sonar dataset, as shown in Fig. 5. From Fig. 5, we can see that A_FC using an adaptive strategy has a better population diversity than S_FC at the 10th and 30th generation. Moreover, A_FC shows better fitness values than S_FC. The experimental results confirm

Table 3 continued

Dataset	Method	#F1	A±Std-KNN	Wtest-KNN	#F2	A±Std-DT	Wtest-DT
Leukemia	FULL	7129	74.26 ± 6.17	++	7129	67.78 ± 17.59	++
	S_FC	1	82.00 ± 9.97	+	1	85.17 ± 14.40	+
	A_FC	1	83.17 ± 8.99	+	1	86.00 ± 10.91	+
	M_FC	8.6	86.50 ± 7.21	=	20.47	87.33 ± 10.31	+
	AM_FC	6.57	86.50 ± 6.73		17.47	89.17 ± 9.92	
Lymphoma	FULL	60	80.00 ± 0.00	+	12582	84.44 ± 8.49	++
	S_FC	1	79.50 ± 5.68	=	1	88.83 ± 13.08	+
	A_FC	1	80.00 ± 3.11	=	1	91.17 ± 11.81	+
	M_FC	1.37	81.33 ± 3.64	=	16.47	95.67 ± 5.73	+
	AM_FC	1.27	81.33 ± 3.64		6	96.33 ± 4.82	
MLL	FULL	12582	58.15 ± 12.65	++	12582	57.22 ± 17.64	++
	S_FC	1	63.17 ± 14.40	++	1	63.33 ± 15.99	++
	A_FC	1	65.00 ± 10.65	+	1	65.17 ± 12.22	++
	M_FC	16.47	68.00 ± 14.58	=	25.57	70.83 ± 14.72	+
	AM_FC	12.60	69.67 ± 10.54		30.40	74.50 ± 17.05	
Ovarian	FULL	15154	85.11 ± 4.32	++	15154	93.89 ± 3.24	+
	S_FC	1	95.94 ± 2.91	+	1	94.30 ± 3.96	+
	A_FC	1	96.17 ± 3.22	=	1	95.39 ± 3.79	=
	M_FC	3.17	98.28 ± 1.61	=	12.77	96.88 ± 2.96	=
	AM_FC	18.23	97.42 ± 2.27		14.93	96.50 ± 2.85	
Colon	FULL	2000	65.74 ± 8.25	++	2000	67.78 ± 15.67	+
	S_FC	1	69.17 ± 21.27	++	1	67.83 ± 13.27	+
	A_FC	1	70.67 ± 13.95	++	1	68.33 ± 13.92	+
	M_FC	12.77	73.33 ± 12.99	+	32.76	71.00 ± 12.94	+
	AM_FC	12.2	78.00 ± 10.54		20.27	74.83 ± 10.68	

that our adaptive strategy can improve the diversity of population and prevent GP from falling into local search, thereby improving the classification performance.

5.2.4 Comparisons between FULL, S_FC, A_FC, M_FC and AM_FC

To verify the effectiveness of our proposed AM_FC, we compare AM_FC with FULL, S_FC, A_FC and M_FC. In KNN and DT classifier, the accuracy of AM_FC is higher than FULL on all ten datasets. Similarly, in KNN and DT classifier, the accuracy of AM_FC method is higher than S_FC and A_FC on all ten datasets. As shown in Figs. 3 and 4, AM_FC can achieve significantly better classification performance than S_FC and A_FC.

Our AM_FC applies our proposed adaptive strategies to M_FC. Comparisons between AM_FC and M_FC aim to verify the effectiveness of adaptive strategies on improving elite archive in our multiple feature construction method. As shown in Table 3, for KNN classifier, AM_FC can achieve better classification performance than M_FC on 7 datasets, with comparable performance on 2 datasets and

slightly worse performance on 1 dataset. On the sonar, liver-disorders, MLL and Colon datasets, the accuracy of AM_FC is 2.01%, 4.56%, 1.67%, and 4.67% higher than that of M_FC, respectively. For DT classifiers, the AM_FC method outperformed the M_FC method slightly in 9 out of 10 datasets, especially in the MLL and Colon datasets, which improved by 3.67% and 3.83%, respectively. The experimental results show that the adaptive strategy used in AM_FC can improve the global search ability of GP, and effectively enable GP jump out local optima when the population converges too early, thereby further improving classification performance.

From the experimental results in Table 3, we can see that M_FC preserves a few or dozens of excellent individuals, and in general, AM_FC obtains more individuals than M_FC, which confirms that the elite archive strategy in multiple feature construction methods can preserve an appropriate number of high-quality individuals, and adaptive strategies can increase the diversity of individuals in the population. It also confirms that the introduction of adaptive strategies can increase the diversity of individuals in the population, expand the GP search range, and thus increase the number

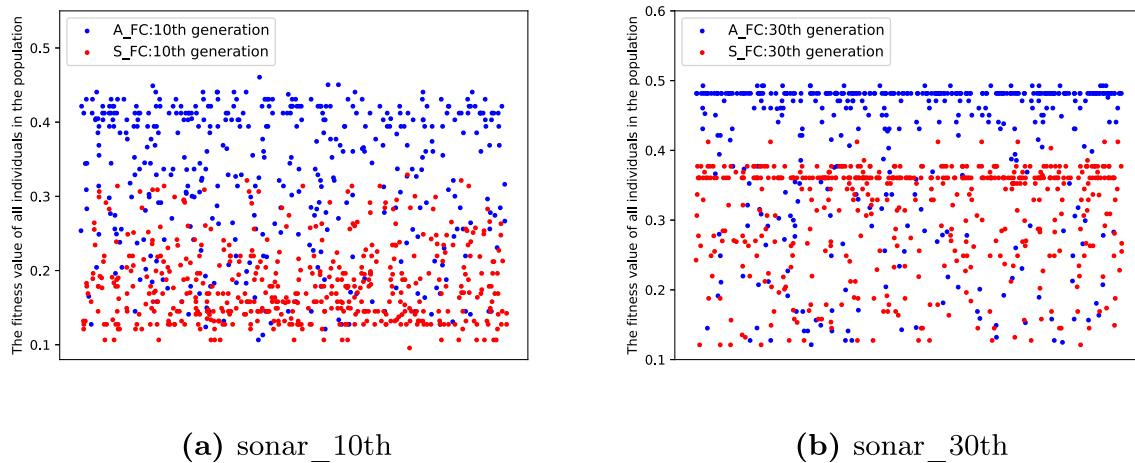


Fig. 5 Scatter plots of population diversity for A_FC and S_FC methods on sonar dataset

of individuals. In order to provide more evidences that adaptive strategies can expand the GP's search space, we obtained the average fitness values with generations for AM_FC and M_FC on datasets sonar, WBC and Ovarian. The curves are shown in Fig. 6. As shown in Fig. 6, the average fitness values of three datasets gradually increase with generations. On three datasets, the average fitness values of AM_FC is consistently higher than that of M_FC. The average fitness values indicates the average quality of individuals in a population, the higher the average fitness values shows GP have evolved more effective features. When performing multiple feature construction, the individuals in the storage pool will be better than the individuals without using the adaptive strategy, and to the quality of the individuals increases ensure that the selected multiple individuals from the storage pool will be more effective for multi-feature construction.

5.3 Comparisons between AM_FC and four benchmark techniques (FcNesh, FcTran, Fcm and FcMa)

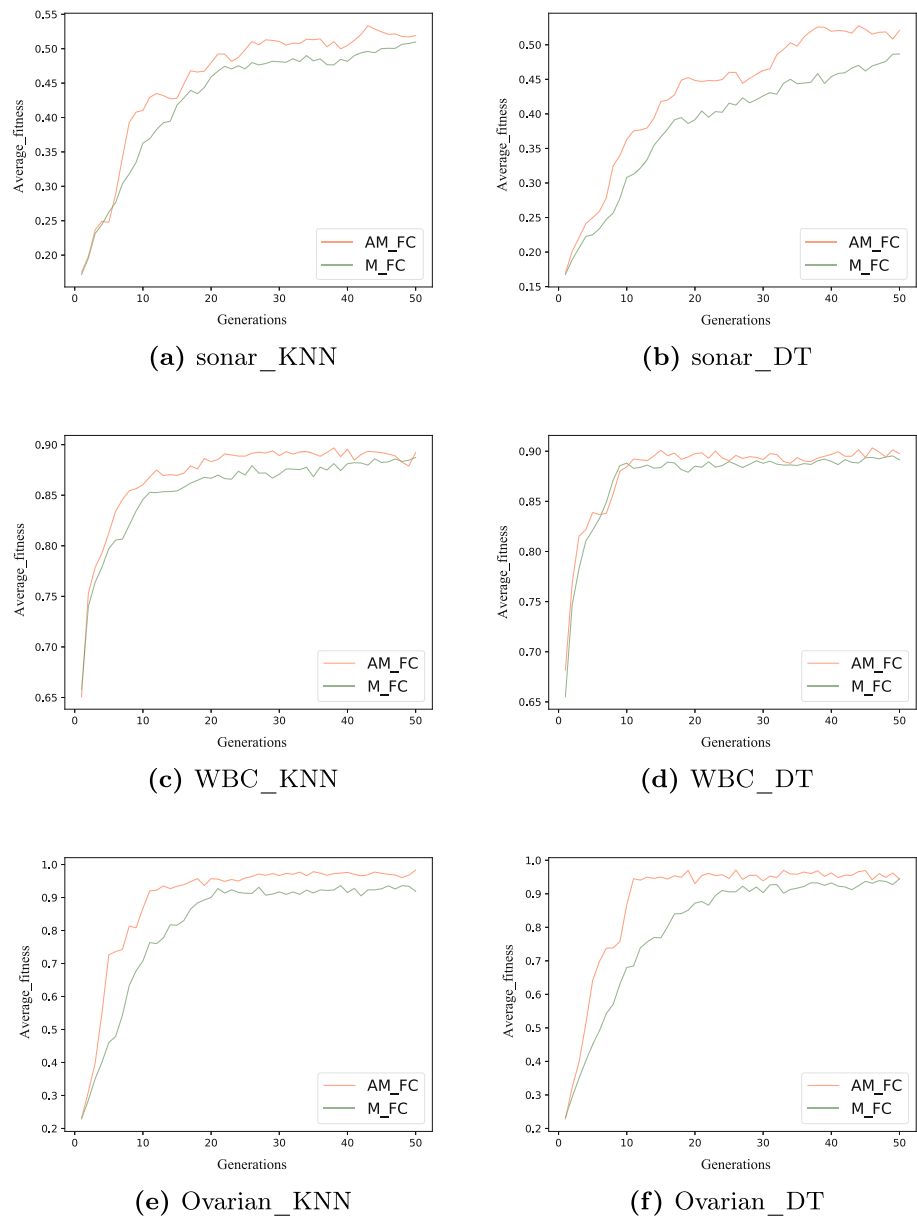
Table 4 lists the experimental results of our proposed method AM_FC and other four benchmarks. In the table, “#F1” denotes the number of constructed features using the KNN classifier. “#F2” represents the number of constructed features using the DT classifier. “A $\pm$ Std-KNN” and “A $\pm$ Std-DT” denote the average accuracy after 30 independent runs using the KNN and DT classification algorithms, respectively, as well as the corresponding standard deviation. The best experimental results for each dataset on the four methods are marked in bold. The meaning of the above attribute names is the same in Table 4.

As shown in Table 4, for KNN classifier, the classification performance of our proposed method AM_FC is higher than the other four methods in 8 out of 10 datasets, and the

classification performance of AM_FC is consistent with the classification performance of FcMa in Lymphoma dataset, slightly lower than FcTran. On the Ovarian dataset, it is only slightly lower than FcTran. The classification accuracy of AM_FC is 10.95%, 10.35%, 6.38% and 6.54% higher than that of FcNesh, FcTran, Fcm and FcMa on liver-disorders dataset, 19.67%, 1.24%, 0.84% and 3.17% higher than that of FcNesh, FcTran, Fcm and FcMa on MLL dataset, and higher than FcNesh, FcTran and FcMa by 13.74%, 12.17%, 8.33% and 8.9% on Colon dataset. Similarly, for DT classifier, the AM_FC method obtains the highest classification performance on 10 datasets. For example, AM_FC is 6.75%, 6.65%, 6.54% and 7.51% higher than FcNesh, FcTran, Fcm and FcMa on sonar dataset, and 5.52%, 2.04%, 5.26% and 7.17% higher than FcNesh, FcTran, Fcm and FcMa on liver-disorders dataset, respectively. It is 10.98%, 18.22%, 6.02% and 4.67% higher than FcNesh, FcTran, Fcm and FcMa on MLL dataset, and 11.13%, 6.72%, 4.36% and 3.5% higher than FcNesh, FcTran, Fcm and FcMa on Colon dataset, respectively. The experimental results show that our proposed AM_FC shows better classification performance compared with other four benchmarks.

To avoid the unfair comparison with other methods caused by adaptive strategy, we can compare our multiple feature construction method M_FC without adaptive strategy with benchmark techniques. Under KNN classifier, M_FC can obtain better classification results than the other four methods on 8 out of 10 datasets. For example, the classification accuracy of M_FC is 10.2%, 1.79%, 3.66% and 4.33% higher than that of FcNesh, FcTran, Fcm and FcMa on Leukemia dataset. Similarly, for DT classifier, M_FC can achieve better classification performance than the other four methods on 8 out of 10 datasets. The experimental results show that our proposed method M_FC still has advantages compared to benchmark techniques.

Fig. 6 The curve of average fitness values with generations in AM_FC and M_FC



5.4 Further discussions

This paper investigates how to improve the performance of feature construction methods. We propose an elite archive-based multiple feature construction methods, which utilizes adaptive strategies to preserve effective discriminative construction features during GP evolution. Experiments have verified the effectiveness of our proposed methods and we summarize the following points.

5.4.1 Feature construction methods

GP-based feature construction is an optional way to transform the original features into another feature space and

obtain more effective features. GP can discover the hidden relationships between features, establish the relationships between features to construct new features, and sometimes the new constructed features can achieve very good classification accuracy. For example, on the gene microarray datasets experimented in this paper, feature construction often achieves high experimental results. In addition, feature construction methods only need to convert thousands of original features into one or several constructed features, thus achieving the goal of feature dimensionality reduction. The effectiveness of feature construction methods is dataset dependent. Our previous experiments have found that feature construction can often achieve higher classification performance than original features on datasets with relationships

Table 4 The experimental results of FcNesh, FcTran, Fcm, FcMa and AM_FC

Dataset	Method	#F1	A±Std-KNN	#F2	A±Std-DT
sonar	FcNesh	2	65.67 ± 9.39	2	66.94 ± 8.58
	FcTran	12.97	65.72 ± 7.57	27	67.42 ± 8.95
	Fcm	10	66.50 ± 6.94	10	66.95 ± 7.45
	FcMa	33.5	68.90 ± 6.89	18.27	66.45 ± 7.48
	AM_FC	20.87	74.32 ± 5.42	43.93	73.96 ± 4.32
wdbc	FcNesh	2	92.54 ± 4.56	2	94.78 ± 2.48
	FcTran	21.33	96.02 ± 1.87	20.83	93.97 ± 1.97
	Fcm	10	95.31 ± 1.66	10	94.17 ± 2.09
	FcMa	5.6	95.83 ± 1.43	16.5	94.91 ± 2.38
	AM_FC	10.07	97.08 ± 1.30	32.77	95.80 ± 2.05
iris	FcNesh	3	95.11 ± 3.22	3	92.67 ± 2.82
	FcTran	4.63	93.87 ± 4.16	4.87	94.48 ± 4.83
	Fcm	10	93.97 ± 5.01	10	94.56 ± 4.25
	FcMa	2.3	94.02 ± 4.33	2.37	94.22 ± 4.83
	AM_FC	11.4	95.72 ± 3.54	16.57	95.68 ± 2.96
liver-disorders	FcNesh	2	58.79 ± 4.70	2	58.63 ± 5.30
	FcTran	6.9	59.39 ± 6.56	6.97	62.11 ± 6.61
	Fcm	10	63.36 ± 6.41	10	58.89 ± 5.37
	FcMa	15.7	63.20 ± 5.91	15.37	56.98 ± 4.90
	AM_FC	30.8	69.74 ± 5.71	40.43	64.15 ± 5.79
WBC	FcNesh	2	77.75 ± 9.92	2	79.58 ± 10.76
	FcTran	9.23	95.52 ± 1.07	9.0	94.47 ± 1.68
	Fcm	10	95.78 ± 1.32	10	94.68 ± 1.37
	FcMa	7.97	95.98 ± 1.08	12.07	94.67 ± 1.99
	AM_FC	22.5	96.86 ± 1.17	28.30	95.57 ± 1.55

between features, such as on Hillvalley and Balance-scale datasets from UCI Machine Learning Repository. Moreover, the effectiveness of feature construction methods is classifier dependent. Our previous experiments [11] have found that feature construction can achieve good classification performance on classifiers such as KNN, Decision tree related classifiers, Neive Bayes etc.

5.4.2 Multiple feature construction methods

We use IGR as the fitness function, employ elite archive strategy to preserve top excellent individuals, and adopt forward search method to select appropriate number of effective constructed features. Experiments show that our multiple feature construction methods can significantly improve the classification performance compared with traditional single feature construction method. We generally obtain several or dozens of effective construction features on ten datasets, avoiding the problem of setting the number of preserved constructed features in our work [9]. Other studies have proposed multiple feature construction methods such as multi-tree representations and cooperative coevolution strategy [19], which are not based on standard GP. Among multiple feature construction

methods based on standard GP, elite archive is an effective strategy to construct multiple features. Improving the evolutionary strategy of GP and using improved elite archive strategy to obtain more effective constructed features may be an interesting work in the future.

5.4.3 Adaptive GP for multiple feature construction methods

We use the adaptive strategy on GP to adjust the crossover and mutation probabilities with the fitness values of individuals in the process of population evolution. Experiments show that our proposed AM_FC using adaptive GP can improve or maintain the classification performance compared with M_FC without using adaptive GP. Experiments show that AM_FC can obtain more abundant and more effective features, which confirms that adaptive GP can expand the search space of GP, assigning new crossover and mutation probabilities to individuals in each generation, in order to generate new individuals and improve the diversity of the population. Although we have obtained preliminary experimental results, how to improve the adaptive strategy on GP is still a problem that needs to be studied. Some other work such as adjusting

Table 4 continued

Dataset	Method	#F1	A±Std-KNN	#F2	A±Std-DT
Leukemia	FcNesh	2	76.30 ± 9.83	2	80.19 ± 11.89
	FcTran	4	84.71 ± 8.46	35.8	78.11 ± 16.07
	Fcm	10	83.67 ± 11.32	10	85.96 ± 8.79
	FcMa	5.6	82.17 ± 8.23	6.17	86.50 ± 9.41
	AM_FC	6.57	86.50 ± 6.73	17.47	89.17 ± 9.92
Lymphoma	FcNesh	3	78.89 ± 2.48	3	78.22 ± 11.79
	FcTran	2.48	82.11 ± 2.85	20.6	91.33 ± 7.18
	Fcm	10	81.33 ± 3.64	10	94.50 ± 6.53
	FcMa	2.3	81.33 ± 3.64	5	96.17 ± 6.41
	AM_FC	1.27	81.33 ± 3.64	6	96.33 ± 4.82
MLL	FcNesh	3	50.00 ± 16.29	3	63.52 ± 13.04
	FcTran	10.2	68.43 ± 12.09	70.37	56.22 ± 18.90
	Fcm	10	68.83 ± 12.16	10	68.48 ± 18.75
	FcMa	9.53	66.50 ± 14.38	9.77	69.83 ± 15.67
	AM_FC	13.23	69.67 ± 16.78	30.40	74.50 ± 17.05
Ovarian	FcNesh	2	95.42 ± 5.10	2	95.02 ± 4.20
	FcTran	4	97.86 ± 1.22	38.57	94.64 ± 5.18
	Fcm	10	96.33 ± 2.47	10	96.20 ± 1.56
	FcMa	10.17	95.83 ± 1.43	9.53	96.35 ± 2.39
	AM_FC	18.23	97.42 ± 2.27	14.93	96.50 ± 2.85
Colon	FcNesh	2	64.26 ± 11.08	2	63.70 ± 10.90
	FcTran	12.97	65.83 ± 12.45	66.1	68.11 ± 16.14
	Fcm	10	69.67 ± 13.84	10	70.47 ± 12.67
	FcMa	6.8	69.10 ± 16.50	7.67	71.33 ± 13.90
	AM_FC	12.2	78.00 ± 10.54	20.27	74.83 ± 10.68

parameters in P_m and P_c , rationally utilizing the population size and using different adaptive strategies in different evolutionary periods to improve the search performance, also need to be investigated.

6 Conclusions and future work

This paper proposes a multiple feature construction method that applies an elite archive strategy to GP to preserve and select effective constructed features in the storage pool, and employs an adaptive strategy to prevent GP from converging too early and generate more effective constructed features. Experiments show that compared with the single feature construction method, our proposed multiple feature construction method can significantly improve the classification performance. Our archive-based multiple feature construction methods using adaptive GP can maintain or further improve the classification performance compared with the method without using adaptive GP. Comparisons with four state-of-the-art techniques show that our proposed method has achieved significantly better results.

Though our proposed AM_FC method shows good classification results, there are still some issues that need to be solved. Firstly, our work has not addressed the issues of GP's bloat problem [37], which can make GP individuals too complex and lead to overfitting problems. The multi-objective [38, 39] strategy can be introduced to control the size of GP individuals. Secondly, we only choose IGR as the fitness function, and there are many other statistical measures that have not been experimented. In the future, we will focus on employing or improving other fitness functions, and further improve the effectiveness of our proposed AM_FC method. Moreover, due to the need for real-time monitoring of fitness changes and archiving elite individuals in our work, there is an increase in computational cost. High-dimensional datasets affect GP's search effect, thereby reducing classification performance. Investigating other strategies to reduce computational consumption and improve the classification performance of high-dimensional datasets is our future research work.

Author Contributions Jia and Ma wrote the main manuscript text and prepared figures 1-9. Zhang and Gao mainly did corrective revisions of thesis papers. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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